

PDE-based Model for Nitrifying Bacteria

Parameter Non-dimensionalize

References $\rightarrow L = 5 \times 10^{-4} \text{ m}$ $C_{\text{ref}} = C_{\text{O}, \text{NH}_4}$
 $u_{\text{ref}} = \frac{P_{\text{packed}}}{M_{\text{biomass}}} \approx 2.03 \times 10^7 \mu\text{M} \approx 20.3 \text{ M}$

Independent and state variables

$$\hat{x} = x/L \quad \hat{t} = t/r_{\text{max}} \quad \hat{r} = r/L$$
$$\hat{u}_i = u_i/u_{\text{ref}} \quad \hat{g}_j = g_j/C_{\text{ref}} \quad \hat{p} = \sum \hat{u}_i = p/u_{\text{ref}}$$

$$\{r_{\text{max}} = \max(r_i)\}$$

$$\tau_{\text{slow}} = 1/r_{\text{max}} \quad \tau_{\text{fast}} = L^2/D_{\text{NH}_4}$$

$$\hat{r}_i = r_i/r_{\text{max}} \quad \hat{b}_i = b_i/r_{\text{max}} \quad \hat{K}_{ij} = K_{ij}/C_{\text{ref}}$$

Coefficients $\hat{D}_i = d_i/r_{\text{max}} L^2$ $\hat{A}_i = a_i/r_{\text{max}} L^2$

$\hat{A}_i/\hat{D}_i = a_i/d_i$ ratio is preserved under this scaling.

Substrate-specific dimensionless terms

$$E_j = r_{\text{max}} L^2/D_j \quad \varepsilon = \max_j E_j \quad \delta_i = d_i/D_{\text{ref}}$$

Damköhler Number

$$Da_{ij} = u_{\text{ref}}/Y_{ij} C_{\text{ref}}$$

↳ ratio of how much biomass the reference substrate scale can support ideally to actual reference biomass scale.
 $\rightarrow$ Decides if eqⁿ is diffusion-driven or rxn-driven.

Elliptic Eqⁿ coefficients

$$\lambda_{ij} = \frac{L^2 u_{\text{ref}} r_i}{D_j C_{\text{ref}} Y_{ij}} \quad \text{decomposes as: } \lambda_{ij} = \hat{r}_i E_j \cdot Da_{ij}$$

$\rightarrow \{\text{consumption}\}$

Production $\rightarrow \lambda_i^{\text{prod}} = L^2 \cdot u_{\text{ref}} \cdot \hat{r}_i / D_{\text{prod}} C_{\text{ref}} Y_{i, \text{source}}$

Boundary Feed values $\rightarrow \hat{C}_{\text{O}, j} = C_{\text{O}, j}/C_{\text{ref}}$

Bacterial Eqs

$$\frac{\partial \hat{u}_{AOB}}{\partial \hat{t}} = \hat{D}_{AOB} \nabla^2 \hat{u}_{AOB} + \hat{A}_{AOB} \nabla \cdot (\hat{u}_{AOB} \nabla \hat{p}) + \hat{u}_{AOB} \left[\hat{r}_{AOB} \left(M(\hat{c}_{NH_4}; \hat{K}_{AOB, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{AOB, O_2}) - \hat{b}_{AOB} \hat{p} \right) \right]$$

$$\frac{\partial \hat{u}_{NOB}}{\partial \hat{t}} = \hat{D}_{NOB} \nabla^2 \hat{u}_{NOB} + \hat{A}_{NOB} \nabla \cdot (\hat{u}_{NOB} \nabla \hat{p}) + \hat{u}_{NOB} \left[\hat{r}_{NOB} M(\hat{c}_{NO_2}; \hat{K}_{NOB, NO_2}) M(\hat{c}_{O_2}; \hat{K}_{NOB, O_2}) - \hat{b}_{NOB} \hat{p} \right]$$

$$\frac{\partial \hat{u}_{CMX}}{\partial \hat{t}} = \hat{D}_{CMX} \nabla^2 \hat{u}_{CMX} + \hat{A}_{CMX} \nabla \cdot (\hat{u}_{CMX} \nabla \hat{p}) + \hat{u}_{CMX} \left[\hat{r}_{CMX} M(\hat{c}_{NH_4}; \hat{K}_{CMX, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{CMX, O_2}) - \hat{b}_{CMX} \hat{p} \right]$$

$$\hat{p} = \hat{u}_{AOB} + \hat{u}_{NOB} + \hat{u}_{CMX}$$

PDE-Model for Nitrifying Bacteria

$$M(c; K) := \frac{c}{K + c} \quad (1)$$

$$0 = \nabla^2 \hat{c}_{NH_4} - \Lambda_{AOB, NH_4} \hat{u}_{AOB} M(\hat{c}_{NH_4}; \hat{K}_{AOB, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{AOB, O_2}) - \Lambda_{CMX, NH_4} \hat{u}_{CMX} M(\hat{c}_{NH_4}; \hat{K}_{CMX, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{CMX, O_2}) \quad (2)$$

$$0 = \nabla^2 \hat{c}_{NO_2} + \Lambda_{AOB}^{prod} \hat{u}_{AOB} M(\hat{c}_{NH_4}; \hat{K}_{AOB, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{AOB, O_2}) - \Lambda_{NOB, NO_2} \hat{u}_{NOB} M(\hat{c}_{NO_2}; \hat{K}_{NOB, NO_2}) M(\hat{c}_{O_2}; \hat{K}_{NOB, O_2}) \quad (3)$$

$$0 = \nabla^2 \hat{c}_{NO_3} + \Lambda_{NOB}^{prod} \hat{u}_{NOB} M(\hat{c}_{NO_2}; \hat{K}_{NOB, NO_2}) M(\hat{c}_{O_2}; \hat{K}_{NOB, O_2}) + \Lambda_{CMX}^{prod} \hat{u}_{CMX} M(\hat{c}_{NH_4}; \hat{K}_{CMX, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{CMX, O_2}) \quad (4)$$

$$0 = \nabla^2 \hat{c}_{O_2} - \Lambda_{AOB, O_2} \hat{u}_{AOB} M(\hat{c}_{NH_4}; \hat{K}_{AOB, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{AOB, O_2}) - \Lambda_{NOB, O_2} \hat{u}_{NOB} M(\hat{c}_{NO_2}; \hat{K}_{NOB, NO_2}) M(\hat{c}_{O_2}; \hat{K}_{NOB, O_2}) - \Lambda_{CMX, O_2} \hat{u}_{CMX} M(\hat{c}_{NH_4}; \hat{K}_{CMX, NH_4}) M(\hat{c}_{O_2}; \hat{K}_{CMX, O_2}) \quad (5)$$